

Chapter 4

The Common Base Amplifier

Core Topology, Impedances, Base Resistance, Biasing, & CE/CB Duality

*Based on Razavi, Fundamentals of Microelectronics, 2nd Ed.
Chapter 5.3.2 (Common-Base Stage)*

With insights from Razavi's BJT YouTube Lectures (25–27)

Topics Covered:

- Review: the three BJT amplifier topologies
- Physical intuition of the Common Base (CB) stage
- Small-signal voltage gain (positive!)
- Input impedance: why $1/g_m$ is the defining feature
- Output impedance and the CE/CB equivalence
- Effect of source resistance: the $1/(1 + g_m R_S)$ attenuation
- Effect of base resistance and impedance reflection
- Output impedance with Early effect (degeneration duality)
- Biasing the CB stage: coupling and bypass capacitors
- Complete CE vs. CB comparison and design trade-offs
- Practice problems and common mistakes

Chapter 4: Quick Reference Formula Sheet

Basic CB Stage (No R_B , $V_A = \infty$)

Parameter	Formula	Notes
Voltage gain	$A_v = g_m R_C$	Positive (non-inverting)
Input impedance	$R_{in} = \frac{1}{g_m}$	Very low ($\sim 26 \Omega$ at 1 mA)
Output impedance	$R_{out} = R_C$	Same as CE (no r_O)
Current gain	$A_I \approx 1$	$I_C \approx I_E$ (vs. β for CE)

CB Stage with Source Resistance (R_S)

Parameter	Formula	Notes
Overall gain	$A_v = \frac{g_m R_C}{1 + g_m R_S}$	Identical form to CE with R_E
Strong degen.	$A_v \approx R_C / R_S$	If $g_m R_S \gg 1$

CB Stage with Base Resistance (R_B)

Parameter	Formula	Notes
Voltage gain	$A_v = \frac{R_C}{R_E + \frac{R_B}{\beta+1} + \frac{1}{g_m}}$	R_B divided by $\beta + 1$
Input impedance	$R_{in} = \frac{1}{g_m} + \frac{R_B}{\beta + 1}$	Reflection from base to emitter

CB Output Impedance with Early Effect

Parameter	Formula	Notes
Transistor R_{out}	$[1 + g_m(R_E \parallel r_\pi)]r_O$	Same as CE degeneration!
Circuit R_{out}	$R_{out,transistor} \parallel R_C$	R_C still in parallel

1 Review: The Three BJT Amplifier Topologies

In Chapter 3, we mastered the Common Emitter (CE) stage. We now study the second of the three fundamental BJT amplifier topologies:

The Three BJT Topologies at a Glance

Topology	Input	Output	Common	Gain
CE (Ch. 3)	Base	Collector	Emitter	$-g_m R_C$
CB (Ch. 4)	Emitter	Collector	Base	$+g_m R_C$
CC (Ch. 5)	Base	Emitter	Collector	≈ 1

The name “Common Base” means the base terminal is *common* (shared, at AC ground) between the input loop and the output loop.

CONCEPT: Why Study the CB Stage?

At first glance, the CB stage seems inferior to the CE stage: it has the same voltage gain magnitude but a much lower input impedance and only unity current gain. So why bother?

- **High-frequency performance:** The CB stage avoids the “Miller Effect” capacitance that cripples the CE’s bandwidth. This makes it essential in RF and high-speed circuits.
- **Current sensing:** The low $1/g_m$ input impedance is ideal for converting current signals to voltage (transimpedance).
- **Impedance matching:** A $50\ \Omega$ input impedance perfectly matches RF transmission lines.
- **Cascode building block:** The CB stage stacked on a CE stage forms the **cascode** — one of the most important circuits in analog IC design.

LECTURE INSIGHT (Lecture 25)

Razavi introduces the CB stage: “The common-base topology may seem like a step backward — same gain magnitude, worse input impedance. But later, when we study high-frequency behavior, you’ll see that the CB stage is fundamentally superior to CE at high speeds. It’s also the key building block for the cascode, which combines the best of both worlds.”

2 The Basic Common Base Stage

2.1 Circuit Topology and Physical Intuition

the core and subsequently add the biasing elements.

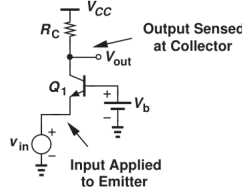


Figure 5.60 Common-base stage

Figure 1: Figure 5.60 from Razavi — The common-base stage. The base is tied to a constant voltage V_b (AC ground), the input is applied at the emitter, and the output is sensed at the collector.

Physical intuition — step by step:

1. Imagine applying a small positive change $+\Delta V$ to the emitter.
2. Since the base voltage is *fixed* (at V_b), the base-emitter voltage $V_{BE} = V_B - V_E$ **decreases**.
3. A *smaller* V_{BE} means *less* collector current: $\Delta I_C < 0$.
4. With less current through R_C , the voltage drop $I_C R_C$ decreases.
5. Therefore $V_{out} = V_{CC} - I_C R_C$ **increases**.
6. **Result:** Input goes up $\rightarrow$ output goes up. The CB stage is **non-inverting**.

KEY RESULT:

The CB stage has a **positive voltage gain**. Unlike the CE stage, the output moves in the **same direction** as the input. This is because the signal enters at the emitter (the “bottom” of V_{BE}), so increasing V_{in} *decreases* V_{BE} rather than increasing it.

2.2 Large-Signal Behavior

Chapter 5 Bipolar Amplifiers

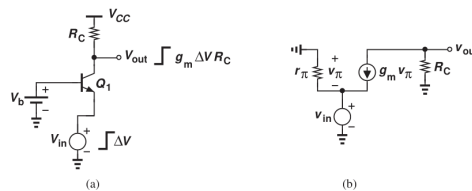


Figure 5.61 (a) Response of CB stage to small-signal change. (b) small-signal model

Figure 2: Figure 5.61 from Razavi — (a) Response of the CB stage to an input change. (b) I/O characteristic showing the non-inverting behavior. Note the voltage headroom constraint is similar to CE.

The I/O characteristic mirrors that of the CE stage, except it is **not flipped**. As V_{in} increases from 0 to V_b :

- For small V_{in} : V_{BE} is large, I_C is high, and V_{out} is low (near $V_{CE,sat}$).
- For V_{in} near V_b : V_{BE} vanishes, $I_C \rightarrow 0$, and $V_{out} \rightarrow V_{CC}$.
- The active region is the steep middle portion with slope $= +g_m R_C$.

Similar to the CE stage limitation, this condition translates to

$$A_v = \frac{I_C}{V_T} \cdot R_C \quad (5.255)$$

$$= \frac{V_{CC} - V_{BE}}{V_T} \quad (5.256)$$

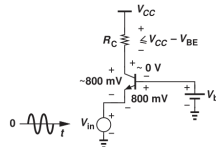


Figure 5.60 Voltage headroom in CB stage

Figure 3: Figure 5.62 from Razavi — Voltage headroom in the CB stage. The output swing is constrained between $V_{CE,sat}$ and V_{CC} .

2.3 Small-Signal Voltage Gain

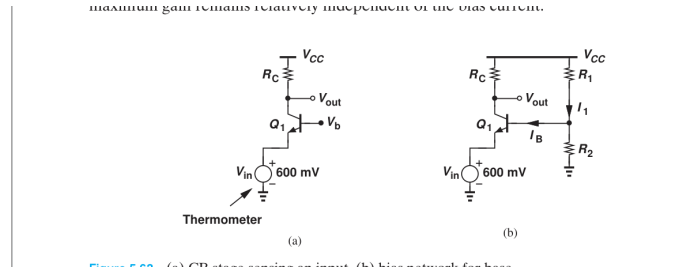


Figure 4: Figure 5.63 from Razavi — (a) CB stage sensing an input voltage through a source resistance R_S . (b) Small-signal equivalent circuit.

DERIVATION: CB Stage Voltage Gain

Replace the transistor with its small-signal model (with $V_A = \infty$). Since the base is grounded and v_{in} is applied to the emitter:

$$v_{\pi} = v_B - v_E = 0 - v_{in} = -v_{in}$$

The controlled current source produces $g_m v_{\pi} = -g_m v_{in}$ flowing downward through the collector. This current flows upward through R_C (from ground):

$$v_{out} = -(-g_m v_{in}) \cdot R_C = g_m v_{in} R_C$$

Therefore:

$$A_v = \frac{v_{out}}{v_{in}} = +g_m R_C$$

The gain has the same magnitude as the CE stage but is positive (non-inverting).

EXAM TIP

Quick memory aid: CE inverts ($-g_m R_C$), CB does not ($+g_m R_C$). The reason: in CE, an increase in v_{in} *increases* V_{BE} ; in CB, an increase in v_{in} *decreases* V_{BE} (it enters from the other side!). This sign flip cascades through the entire analysis.

3 Input Impedance: The Defining Feature of CB

3.1 Derivation

The most important and distinctive feature of the CB stage is its extremely low input impedance.

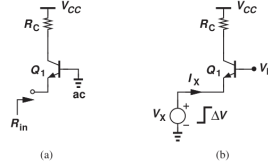


Figure 5.64 (a) Input impedance of CB stage. (b) Equivalent small-signal model.

Figure 5: Figure 5.64 from Razavi — (a) Input impedance of the CB stage. (b) Small-signal model showing the test voltage applied at the emitter with the base grounded.

DERIVATION: CB Input Impedance

Apply a test voltage v_X at the emitter with the base grounded. Then $v_\pi = 0 - v_X = -v_X$. KCL at the emitter node (all currents leaving the node):

$$i_X = \frac{v_\pi}{r_\pi} + g_m v_\pi = v_\pi \left(\frac{1}{r_\pi} + g_m \right)$$

Substituting $v_\pi = -v_X$:

$$i_X = -(-v_X) \left(\frac{1}{r_\pi} + g_m \right) = v_X \left(\frac{1}{r_\pi} + g_m \right)$$

Since $1/r_\pi = g_m/\beta \ll g_m$ (for $\beta \gg 1$):

$$R_{in} = \frac{v_X}{i_X} = \frac{1}{g_m + 1/r_\pi} \approx \frac{1}{g_m}$$

Physical intuition: In the CE stage (looking into the base), you see $r_\pi = \beta/g_m$ because only $1/(\beta + 1)$ of the emitter current enters the base. In the CB stage (looking into the emitter), you see $1/g_m$ because the emitter carries the *full* current. This is a factor of β smaller!

CONCEPT: $1/g_m$ — The Universal “Emitter Resistance”

Whenever you look into the emitter of a BJT (with the base at AC ground), you see $1/g_m$. This appears in many contexts:

- CB input impedance: $R_{in} = 1/g_m$
- Emitter Follower output impedance: $R_{out} \approx 1/g_m$ (Chapter 5)
- Degeneration denominator: the $1/g_m$ term in $A_v = -R_C/(1/g_m + R_E)$

For $I_C = 1$ mA: $1/g_m = V_T/I_C = 26$ mV/1 mA = 26 Ω . Very small!

3.2 Implications of Low Input Impedance

The low $R_{in} = 1/g_m$ has profound consequences:

- **Advantage:** Ideal for $50\ \Omega$ impedance matching in RF circuits and transmission lines.
- **Advantage:** Works well as a current-to-voltage converter (transimpedance amplifier).
- **Disadvantage:** Any source resistance R_S creates massive signal attenuation through voltage division.

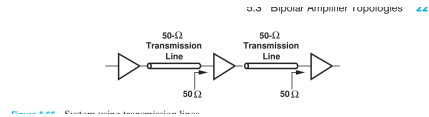


Figure 6: Figure 5.65 from Razavi — A CB stage used to receive a signal from a transmission line. The $1/g_m$ input impedance matches the $50\ \Omega$ characteristic impedance of the line.

4 Output Impedance

4.1 Without Early Effect

Figure 5.66 System using transmission lines.

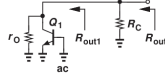


Figure 5.66 Output impedance of CB stage.

Figure 7: Figure 5.66 from Razavi — Output impedance of the CB stage (looking into the collector node).

Setting $v_{in} = 0$ (emitter grounded): $v_{\pi} = 0$, the controlled source $g_m v_{\pi} = 0$ is off. The test current at the collector flows only through R_C :

$$R_{out} = R_C \quad (\text{without } r_O)$$

This is identical to the CE stage. In fact:

■ KEY RESULT:

The output impedance of the CB stage equals that of the CE stage under identical conditions (V_A , degeneration). This must be true: when computing output impedance, we set the input to zero. With $v_{in} = 0$, we have no knowledge of where the input terminal was — the circuit looks the same from the collector regardless of whether it was a CE or CB stage!

4.2 With Early Effect (r_O)

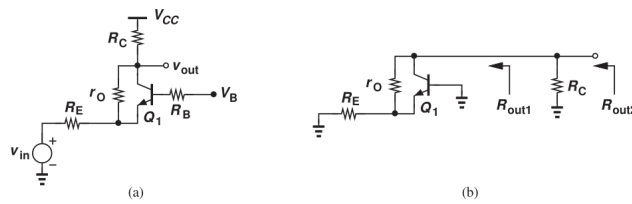


Figure 5.69 (a) General CB stage. (b) Output impedance seen at different nodes.

Figure 8: Figure 5.69 from Razavi — (a) General CB stage with R_E , R_B , and r_O . (b) Output impedance seen at different nodes: R_{out1} at the transistor collector, R_{out2} at the circuit output.

When r_O is included and R_E is present in the emitter, the output impedance looking into the collector of the transistor is the same as the emitter-degenerated CE stage:

$$R_{out1} = [1 + g_m(R_E \parallel r_{\pi})]r_O + (R_E \parallel r_{\pi})$$

The circuit output is:

$$R_{out} = R_{out1} \parallel R_C$$

250 Chapter 5 Bipolar Amplifiers

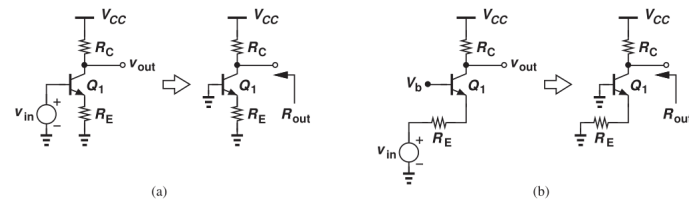


Figure 9: Figure 5.70 from Razavi — (a) CE stage and (b) CB stage simplified for output impedance calculation. With $v_{in} = 0$ in both cases, the circuits are indistinguishable from the collector's perspective.

LECTURE INSIGHT (Lecture 26)

Razavi makes the key point: “When you set the input to zero, the circuit doesn’t know whether it was a CE stage or a CB stage. The output impedance depends only on what’s connected to the emitter and the collector — the input terminal is irrelevant. That’s why the output impedances of CE and CB are always equal.”

5 Effect of Source Resistance

5.1 The Voltage Divider Problem

Because $R_{in} = 1/g_m$ is so small, any source resistance R_S forms a devastating voltage divider:

Since the impedance seen looking into the emitter of Q_1 (from the base grounded) is

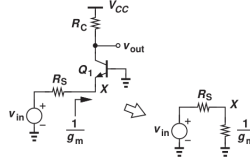


Figure 5.67 CB stage with source resistance

Figure 10: Figure 5.67 from Razavi — CB stage with source resistance R_S . The voltage v_X reaching the emitter is attenuated by the divider between R_S and $1/g_m$.

DERIVATION: CB Gain with Source Resistance

The voltage that actually reaches the emitter through the R_S – R_{in} divider:

$$v_X = v_{in} \cdot \frac{R_{in}}{R_S + R_{in}} = v_{in} \cdot \frac{1/g_m}{R_S + 1/g_m} = \frac{v_{in}}{1 + g_m R_S}$$

Since the CB core gain from emitter to output is $g_m R_C$:

$$v_{out} = g_m R_C \cdot v_X = \frac{g_m R_C}{1 + g_m R_S} \cdot v_{in}$$

$$A_v = \frac{v_{out}}{v_{in}} = \frac{g_m R_C}{1 + g_m R_S}$$

If $g_m R_S \gg 1$: $A_v \approx R_C/R_S$ (gain set by resistor ratio, just like CE degeneration!).

KEY RESULT:

The effect of R_S in a CB stage is **mathematically identical** to emitter degeneration R_E in a CE stage. Both produce the same gain formula: $A_v = g_m R_C / (1 + g_m R_{series})$. From a small-signal perspective, a CB stage with source resistance R_S is equivalent to a degenerated CE stage with $R_E = R_S$!

Example 5.36: CB for 50 Ω Matching (Razavi)

Problem: Design a CB stage to match a 50 Ω source ($R_S = 50 \Omega$). Find the required I_C and the voltage gain if $R_C = 50 \Omega$ (also a 50 Ω load).

Solution:

1. For $R_{in} = 1/g_m = 50 \Omega$: $g_m = 1/(50 \Omega) = 20 \text{ mA/V}$.
2. $I_C = g_m V_T = 20\text{m} \times 26\text{m} = \mathbf{0.52 \text{ mA}}$.
3. Overall gain: $A_v = g_m R_C / (1 + g_m R_S) = 20\text{m} \times 50 / (1 + 20\text{m} \times 50) = 1/2 = \mathbf{0.5}$.

Conclusion: The CB stage cannot provide voltage gain greater than 0.5 when driving a 50 Ω load through a 50 Ω source. However, it provides excellent **power** matching and

low-noise performance, making it ideal for RF receiver front-ends.

EXAM TIP

Current gain of CB = 1: Unlike the CE stage where $A_I = \beta$, the CB stage has $A_I \approx 1$ because $I_C \approx I_E$ (neglecting base current). The current flowing into the emitter simply emerges from the collector. No current amplification occurs!

6 Effect of Base Resistance

6.1 Input Impedance with R_B

In practice, the base is not perfectly at AC ground but has some resistance R_B (from the bias network or parasitic effects). How does this affect the circuit?

Figure 5.73 Comparison of CE and CB stages with base resistance.

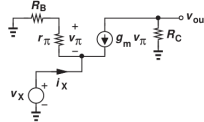


Figure 5.74 Input impedance of CE stage with base resistance.

Figure 11: Figure 5.74 from Razavi — Input impedance of the CE stage in the presence of R_B .

DERIVATION: CB Input Impedance with R_B

From the small-signal model, r_π and R_B form a voltage divider:

$$v_\pi = -\frac{r_\pi}{r_\pi + R_B} v_X$$

KCL at the emitter node:

$$i_X = -\left(\frac{1}{r_\pi} + g_m\right) v_\pi = \left(\frac{1}{r_\pi} + g_m\right) \frac{r_\pi}{r_\pi + R_B} v_X$$

Therefore:

$$R_{in} = \frac{v_X}{i_X} = \frac{r_\pi + R_B}{\beta + 1} \approx \frac{1}{g_m} + \frac{R_B}{\beta + 1}$$

$$R_{in} = \frac{1}{g_m} + \frac{R_B}{\beta + 1}$$

CONCEPT: Impedance Reflection: Base $\rightarrow$ Emitter (Division Rule)

In Chapter 3, we learned that emitter resistance R_E appears **multiplied** by $(\beta + 1)$ when seen from the base. Now we see the **inverse**: base resistance R_B appears **divided** by $(\beta + 1)$ when seen from the emitter.

Looking from Base	Looking from Emitter
R_E appears as $(\beta + 1)R_E$	R_B appears as $R_B/(\beta + 1)$

These are reciprocal rules reflecting the same physics: the base carries $1/(\beta + 1)$ of the emitter current.

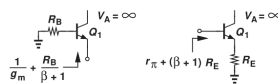


Figure 5.75 Summary of impedance reflection rules.

Figure 12: Figure 5.75 from Razavi — Summary of impedance reflection rules. A resistance at the emitter is multiplied by $(\beta + 1)$ when seen from the base; a resistance at the base is divided by $(\beta + 1)$ when seen from the emitter.

6.2 Voltage Gain with R_B

$$v_P = -\frac{v_{out}}{\beta R_C}(R_B + r_\pi) \quad (5.281)$$

$$= -\frac{v_{out}}{\beta R_C}(R_B + r_\pi). \quad (5.282)$$

We also write a KCL at P :

$$\frac{v_\pi}{r_\pi} + g_m v_\pi = \frac{v_P - v_{in}}{R_E}; \quad (5.283)$$

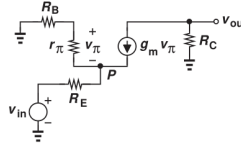


Figure 5.76 CB stage with base resistance

Figure 13: Figure 5.72 from Razavi — Small-signal model of the CB stage with base resistance R_B .

DERIVATION: CB Gain with Base Resistance

The full analysis (with R_E and R_B , $V_A = \infty$) gives:

$$\frac{v_{out}}{v_{in}} = \frac{\beta R_C}{(\beta + 1)R_E + R_B + r_\pi}$$

Dividing numerator and denominator by $(\beta + 1)$:

$$A_v \approx \frac{R_C}{R_E + \frac{R_B}{\beta + 1} + \frac{1}{g_m}}$$

KEY RESULT:

This expression is identical to the CE gain formula with emitter degeneration and base resistance (Eq. 5.185 in Razavi). The only difference is the sign: the CB gain is **positive**. The CE and CB stages are truly dual topologies — same gain magnitude, opposite sign.

Effect of bond wire inductance in a CB stage.

the base, and assuming $V_A = \infty$. From the small-signal equivalent circuit shown in Fig. 5.74, we recognize that

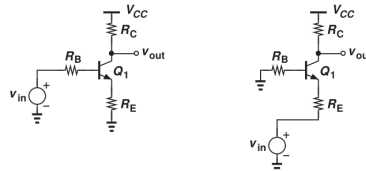


Figure 5.73 Comparison of CE and CB stages with base resistance

Figure 14: Figure 5.73 from Razavi — Comparison showing that CE and CB stages with base resistance have identical gain magnitudes (opposite signs).

LECTURE INSIGHT (Lecture 26)

Razavi emphasizes: “In many students’ minds, the CB stage seems like a completely different circuit from the CE stage. But look at the gain formulas — they’re *identical*! The only difference is the sign. The CE and CB stages are two faces of the same coin. Once you understand one, you understand both.”

7 The “Old Wisdom”: Is CB’s R_{out} Higher Than CE’s?

7.1 A Subtle Misconception

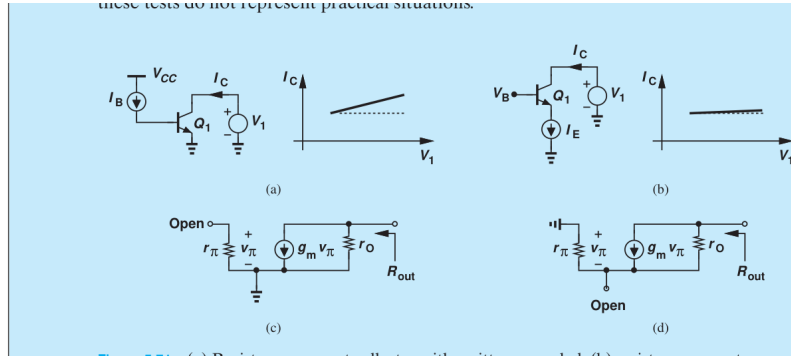


Figure 15: Figure 5.71 from Razavi — (a) I_C vs. V_{CE} with constant base current (slope = $1/r_O$). (b) I_C vs. V_{CE} with constant emitter current (much flatter slope $\sim 1/\beta r_O$). (c), (d) Their small-signal models. These are **not** practical test conditions!

Old textbooks often claim that “the output impedance of the CB stage is much higher than that of the CE stage.” This claim is based on two different experiments:

- **CE test:** Drive the base with a constant *current source* I_B , measure I_C vs. V_{CE} . The slope gives $R_{out} = r_O$.
- **CB test:** Draw a constant *current* from the emitter, measure I_C vs. V_{CE} . The slope gives $R_{out} \approx \beta r_O$.

Example 5.38: Why This Comparison is Misleading (Razavi)

Problem: Explain why the above tests don’t represent practical situations.

Solution:

Both tests use **ideal current sources** to drive the transistor. From a small-signal perspective, current sources are replaced by **open circuits**. So:

- In the CE test [Fig. 5.71(c)]: the base current is zero (open). So $v_{\pi} = 0 \implies g_m v_{\pi} = 0$, and $R_{out} = r_O$.
- In the CB test [Fig. 5.71(d)]: the emitter is open. This is equivalent to infinite emitter degeneration, giving $R_{out} = (1 + g_m r_{\pi}) r_O \approx \beta r_O$.

But in *practice*, neither stage is driven by an ideal current source! Both are driven by finite-impedance voltage sources. Under identical practical conditions (same R_E , same V_A), the output impedances are **equal**.

EXAM TIP

For exams: If asked “which has higher R_{out} , CE or CB?” the answer is: **they are equal** when analyzed under the same conditions (same R_E , same r_O). The “old wisdom” that CB has higher R_{out} comes from comparing apples to oranges (current-source drive vs. voltage drive).

8 Biasing the Common Base Stage

8.1 The DC Path Problem

Biasing a CB stage requires careful attention. The emitter must have a DC path for the bias current, and the base must be at AC ground.

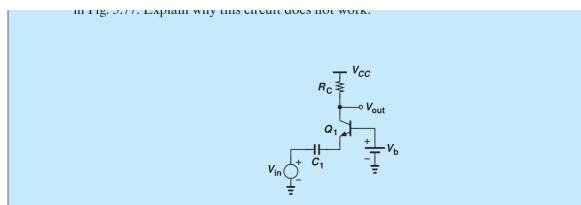


Figure 16: Figure 5.77 from Razavi — CB stage lacking a DC bias current path.

Example 5.40: Why No DC Path Fails (Razavi)

Problem: A student builds a CB stage with no resistor at the emitter, using only a coupling capacitor C_1 (Fig. 5.77). Why doesn't this work?

Solution:

The coupling capacitor blocks DC. Without a resistor from the emitter to ground (or a current source), there is **no DC path** for I_E to flow. The transistor has zero bias current, zero g_m , and zero gain. The circuit is dead.

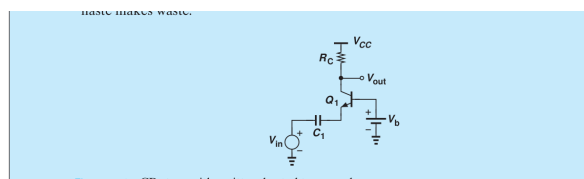


Figure 17: Figure 5.78 from Razavi — CB stage with emitter shorted to ground — the signal is killed.

Example 5.41: Why Shorting the Emitter Fails (Razavi)

Problem: The student then connects the emitter directly to ground (Fig. 5.78). Why doesn't this work either?

Solution:

With the emitter *directly* grounded, $v_E = 0$ regardless of v_{in} . The input signal cannot reach the transistor! The emitter is at AC ground, so $v_{\pi} = 0$ and $v_{out} = 0$. This is the CB equivalent of the CE mistake of shorting the base directly to V_{CC} : the signal has been “shorted out.”

8.2 The Correct Bias Configuration

The correct approach uses:

- R_E : Provides a DC path for I_E and establishes the emitter voltage $V_E = V_b - V_{BE}$.
- C_1 (input coupling cap): Blocks DC from the signal source while allowing the AC signal to reach the emitter.

in Fig. 5.77(b), such a circuit attenuates the signal, lowering the overall voltage gain.

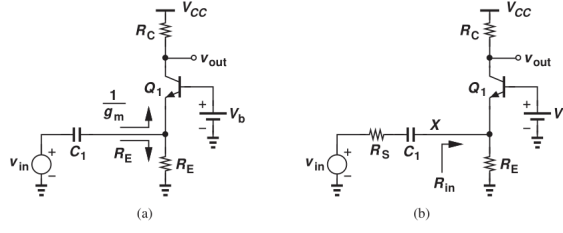


Figure 5.79 (a) CE stage with proper biasing. (b) Inclusion of source resistance.

Figure 18: Figure 5.79 from Razavi — (a) CE stage with proper biasing: R_E provides the DC path, C_1 couples the signal in. (b) With source resistance R_S : the signal is attenuated by the R_S – $(1/g_m \parallel R_E)$ divider.

- R_1, R_2 (voltage divider): Set V_b at the base.
- C_B (base bypass cap): Shorts the base to AC ground at signal frequencies, eliminating the gain degradation from $R_B = R_1 \parallel R_2$.

8.3 Input Impedance with R_E in Parallel

With R_E connected from the emitter to ground, the AC input impedance becomes:

$$R_{in} = \frac{1}{g_m} \parallel R_E$$

Since $R_E \gg 1/g_m$ in a well-designed circuit, $R_{in} \approx 1/g_m$. The condition for R_E to be negligible is:

$$R_E \gg \frac{1}{g_m} \implies I_C R_E \gg V_T$$

Using $R_E \gg 1/g_m$ to ensure the input impedance negligible, we obtain:

$$R_E \gg \frac{1}{g_m} \quad (5.300)$$

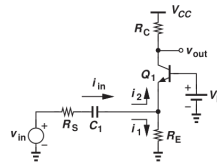


Figure 5.80 Small-signal input current analysis in a CE stage.

Figure 19: Figure 5.80 from Razavi — The input current i_{in} splits between R_E (wasted current i_1) and $1/g_m$ (useful current i_2 that reaches R_C). For $R_E \gg 1/g_m$, most current goes through the transistor.

a short circuit at frequencies of interest [Fig. 5.81(c)].

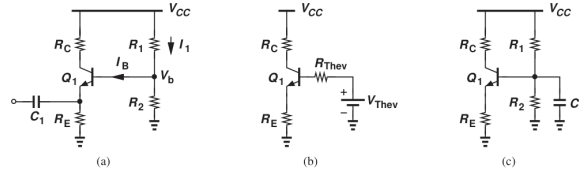


Figure 5.81 (a) CB stage with base bias network. (b) use of Thevenin equivalent.

Figure 20: Figure 5.81 from Razavi — (a) CB stage with R_1/R_2 divider for base biasing. (b) Thevenin equivalent shows $R_{Th} = R_1 \parallel R_2$ in series with the base. (c) Solution: bypass C_B shorts base to ground for AC.

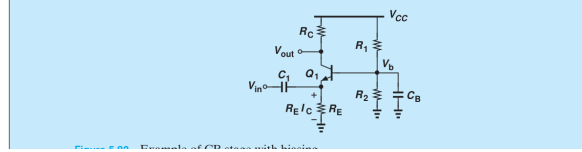


Figure 5.82 Example of CB stage with biasing

Figure 21: Figure 5.82 from Razavi — Complete biased CB stage for Example 5.42.

8.4 Complete Biased CB Stage

Example 5.42: Complete CB Design (Razavi)

Problem: Design a CB stage for $A_v = 10$, $R_{in} = 50 \Omega$. Given: $I_S = 5 \times 10^{-16} \text{ A}$, $V_A = \infty$, $\beta = 100$, $V_{CC} = 2.5 \text{ V}$.

Solution:

- $R_{in} = 1/g_m = 50 \Omega \implies g_m = 20 \text{ mA/V} \implies I_C = g_m V_T = 20\text{m} \times 26\text{m} = \mathbf{0.52 \text{ mA}}$.
- Choose $R_E = 500 \Omega$ ($R_E \gg 1/g_m = 50 \Omega$) to minimize signal shunting.
- With base bypassed: $A_v = g_m R_C = 10 \implies R_C = 10/g_m = \mathbf{500 \Omega}$.
- $V_E = I_E R_E \approx 0.52\text{m} \times 500 = 260 \text{ mV}$.
- $V_{BE} = V_T \ln(I_C/I_S) = 26\text{m} \times \ln(0.52\text{m}/5 \times 10^{-16}) = 26 \times 27.4 = \mathbf{712 \text{ mV}}$.
- $V_b = V_E + V_{BE} = 260 + 712 = \mathbf{972 \text{ mV}}$.
- Divider: choose $I_{div} = 10I_B = 10 \times 0.52\text{m}/100 = 52 \mu\text{A}$.
 $R_1 + R_2 = V_{CC}/I_{div} = 2.5/52\mu = 48.1 \text{ k}\Omega$.
 $R_2/(R_1 + R_2) = V_b/V_{CC} = 972\text{m}/2.5 = 0.389$.
 $R_2 = 0.389 \times 48.1\text{k} = \mathbf{18.7 \text{ k}\Omega}$, $R_1 = \mathbf{29.4 \text{ k}\Omega}$.
- For 900 MHz operation:
 $C_1: |1/(C_1\omega)| \ll 1/g_m \implies C_1 = 20g_m/\omega = 20 \times 20\text{m}/(2\pi \times 900\text{M}) = \mathbf{71 \text{ pF}}$.
 $C_B: |1/(C_B\omega)|/(\beta + 1) \ll 1/g_m \implies C_B = 20(\beta + 1)g_m/((\beta + 1)\omega) = \mathbf{0.7 \text{ pF}}$.

EXAM TIP

Common bypass cap mistake: For the base bypass capacitor C_B , the impedance $|1/(C_B\omega)|$ appears in the gain formula as $R_B/(\beta + 1)$. So C_B only needs to be small enough that its impedance divided by $(\beta + 1)$ is negligible compared to $1/g_m$. Don't waste area on an unnecessarily large C_B !

9 Complete CE vs. CB Comparison

Comprehensive CE vs. CB Comparison

Property	Common Emitter (CE)	Common Base (CB)
Input terminal	Base	Emitter
Common terminal	Emitter (grounded)	Base (AC grounded)
Voltage gain	$-g_m R_C$ (inverting)	$+g_m R_C$ (non-inverting)
Input impedance	$r_\pi = \beta/g_m$ (moderate)	$1/g_m$ (very low)
Output impedance	R_C (or $R_C \parallel r_O$)	R_C (or $R_C \parallel r_O$)
Current gain	β	≈ 1
Effect of R_E	Reduces gain by $1 + g_m R_E$	N/A (equivalent to R_S)
Effect of R_B	Reduces gain by $R_B/(\beta + 1)$	Same — $R_B/(\beta + 1)$
Miller effect	Severe (limits bandwidth)	Absent (high bandwidth)
Best application	Low-freq. voltage amplifier	RF front-end, cascode

CONCEPT: The Cascode: CE + CB = Best of Both Worlds

What if we want the CE's high input impedance *and* the CB's high bandwidth? The answer is the **cascode**: a CE stage driving a CB stage. The CE provides the input impedance (r_π), and the CB eliminates the Miller effect. The result is:

- Input impedance $\approx r_\pi$ (from CE)
- Bandwidth $\approx$ CB bandwidth (Miller-free)
- Output impedance $\approx \beta r_O$ (massively boosted)
- Voltage gain $\approx g_m R_C$ (same as individual stages)

The cascode is studied extensively in more advanced analog courses and is one of the most widely used circuits in integrated circuit design.

10 Practice Problems

Problem 1: Basic CB Analysis

A CB stage has $I_C = 2 \text{ mA}$, $R_C = 1 \text{ k}\Omega$, $\beta = 150$, $V_A = \infty$.

- Find g_m , R_{in} , and A_v .
- If a source with $R_S = 100 \Omega$ drives the stage, find the overall gain.
- What percentage of the signal is “lost” to the voltage divider?

Solution:

(a) $g_m = 2\text{m}/26\text{m} = \mathbf{76.9 \text{ mA/V}}$.

$R_{in} = 1/g_m = \mathbf{13 \Omega}$. $A_v = g_m R_C = 76.9\text{m} \times 1\text{k} = \mathbf{76.9}$.

(b) $A_v = g_m R_C / (1 + g_m R_S) = 76.9 / (1 + 76.9\text{m} \times 100) = 76.9 / 8.69 = \mathbf{8.85}$.

(c) Without R_S : gain = 76.9. With R_S : gain = 8.85. Loss = $(76.9 - 8.85) / 76.9 = \mathbf{88.5\%}$!
The low input impedance causes massive signal attenuation.

Problem 2: CB with Base Resistance

A CB stage has $I_C = 1 \text{ mA}$, $R_C = 2 \text{ k}\Omega$, $\beta = 100$, $R_E = 500 \Omega$, $R_B = R_1 \parallel R_2 = 10 \text{ k}\Omega$. $V_A = \infty$. The base bypass capacitor is absent.

- Find the voltage gain.
- Find the voltage gain with C_B (bypass cap) present.
- By what percentage does R_B degrade the gain?

Solution:

$g_m = 1\text{m}/26\text{m} = 38.5 \text{ mA/V}$. $1/g_m = 26 \Omega$.

(a) Without bypass: $A_v = R_C / (R_E + R_B / (\beta + 1) + 1/g_m) = 2000 / (500 + 10000/101 + 26) = 2000/625 = \mathbf{3.2}$.

(b) With bypass ($R_B \rightarrow 0$): $A_v = R_C / (R_E + 1/g_m) = 2000 / (500 + 26) = \mathbf{3.8}$.

(c) Degradation: $(3.8 - 3.2) / 3.8 = \mathbf{15.8\%}$. The $R_B / (\beta + 1) = 99 \Omega$ term adds significantly to the denominator.

Problem 3: CB Design Problem

Design a CB stage for: $A_v = 20$, $R_{in} = 26 \Omega$, $V_{CC} = 3.3 \text{ V}$, $\beta = 100$, $V_A = \infty$. Find I_C , R_C , R_E , and verify active mode.

Solution:

$R_{in} = 1/g_m = 26 \Omega \implies g_m = 1/(26 \Omega) = 38.5 \text{ mA/V}$.

$I_C = g_m V_T = 38.5\text{m} \times 26\text{m} = \mathbf{1 \text{ mA}}$.

$A_v = g_m R_C = 20 \implies R_C = 20/38.5\text{m} = \mathbf{520 \Omega}$.

Choose $R_E = 500 \Omega (\gg 26 \Omega)$.

$V_E = I_E R_E \approx 1\text{m} \times 500 = 500 \text{ mV}$.

$V_b = V_E + V_{BE} = 500 + 750 \approx 1.25 \text{ V}$.

$V_Y = V_{CC} - I_C R_C = 3.3 - 0.52 = 2.78 \text{ V}$.

$V_{CE} = V_Y - V_E = 2.78 - 0.5 = 2.28 \text{ V} > V_{BE}$. **Active mode confirmed. ✓**

Problem 4: CB Output Impedance

A CB stage has $I_C = 0.5 \text{ mA}$, $R_C = 2 \text{ k}\Omega$, $R_E = 1 \text{ k}\Omega$, $\beta = 100$, $V_A = 10 \text{ V}$.

- Find R_{out} at the transistor collector.
- Find the overall circuit R_{out} .
- Verify this equals the CE degenerated R_{out} .

Solution:

$g_m = 0.5\text{m}/26\text{m} = 19.2 \text{ mA/V}$. $r_\pi = 100/19.2\text{m} = 5.2 \text{ k}\Omega$. $r_O = 10/0.5\text{m} = 20 \text{ k}\Omega$.

(a) $R_E \parallel r_\pi = 1\text{k} \parallel 5.2\text{k} = 839 \Omega$.

$R_{out,transistor} = [1 + g_m(R_E \parallel r_\pi)]r_O = [1 + 19.2\text{m} \times 839] \times 20\text{k} = 17.1 \times 20\text{k} = \mathbf{342 \text{ k}\Omega}$.

(b) $R_{out} = 342\text{k} \parallel 2\text{k} = \mathbf{1.99 \text{ k}\Omega} \approx R_C$ (dominated by R_C).

(c) A CE stage with the same $R_E = 1 \text{ k}\Omega$, $r_\pi = 5.2 \text{ k}\Omega$, $r_O = 20 \text{ k}\Omega$ gives the exact same formula: $R_{out,transistor} = [1 + g_m(R_E \parallel r_\pi)]r_O = 342 \text{ k}\Omega$. **CE and CB output impedances are identical.** ✓

11 Common Mistakes and How to Avoid Them

COMMON MISTAKES

- **Mistake:** Saying the CB gain is $-g_m R_C$ (negative).
Fix: The CB stage is **non-inverting**. $A_v = +g_m R_C$. The input enters at the emitter, so increasing V_{in} decreases V_{BE} , reversing the sign relative to CE.
- **Mistake:** Confusing $R_{in} = r_\pi$ (CE) with $R_{in} = 1/g_m$ (CB).
Fix: Looking into the **base**: $R_{in} = r_\pi = \beta/g_m$. Looking into the **emitter**: $R_{in} = 1/g_m$. The factor of β comes from the current division between base and emitter.
- **Mistake:** Claiming CB has higher R_{out} than CE.
Fix: Under identical conditions (same R_E , same V_A), R_{out} is **identical**. The “old wisdom” compares current-source drive (CB) vs. voltage drive (CE), which is an unfair comparison.
- **Mistake:** Building a CB stage without a DC emitter path.
Fix: The emitter must have a resistor R_E to ground (or a current source) to establish DC bias. Without it, $I_C = 0$ and the amplifier is dead.
- **Mistake:** Shorting the emitter to ground to “provide a DC path.”
Fix: This shorts the AC signal too! Use R_E (with a coupling cap C_1) so that the signal can develop a voltage at the emitter.
- **Mistake:** Making C_B 's impedance small relative to $R_1 \parallel R_2$ instead of $1/g_m$.
Fix: R_B affects the gain through the $R_B/(\beta + 1)$ term. The bypass condition is $|1/(C_B \omega)|/(\beta + 1) \ll 1/g_m$, not $|1/(C_B \omega)| \ll R_1 \parallel R_2$.
- **Mistake:** Ignoring the devastating effect of R_S on CB gain.
Fix: Since $R_{in} = 1/g_m \sim 26 \Omega$, even a modest $R_S = 100 \Omega$ causes $\sim 80\%$ signal attenuation. Always check the source impedance!

Chapter 4: End-of-Chapter Summary

Core CB Stage

- **Topology:** NPN with base at AC ground, input at emitter, output at collector.
- **Gain:** $A_v = +g_m R_C$ (**non-inverting**). Same magnitude as CE.
- **Input impedance:** $R_{in} = 1/g_m$ — very low ($\sim 26\ \Omega$ at 1 mA).
- **Output impedance:** $R_{out} = R_C$ (identical to CE).
- **Current gain:** $A_I \approx 1$ ($I_C \approx I_E$).

With Source and Base Resistance

- **Source resistance R_S :** $A_v = g_m R_C / (1 + g_m R_S)$ — identical to CE degeneration.
- **Base resistance R_B :** $A_v = R_C / (R_E + R_B / (\beta + 1) + 1/g_m)$ — identical (in magnitude) to CE with R_B .
- **Input impedance with R_B :** $R_{in} = 1/g_m + R_B / (\beta + 1)$ — inverse reflection.

Output Impedance (with Early Effect)

- $R_{out,transistor} = [1 + g_m (R_E \parallel r_\pi)] r_O$ — identical to CE with same degeneration.
- CE and CB R_{out} are always equal under the same conditions.

Biasing

- Requires R_E for DC path, C_1 for input coupling, R_1/R_2 divider for V_b , C_B for base bypass.
- $R_E \gg 1/g_m$ to avoid signal shunting.
- C_B must satisfy $|1/(C_B \omega)| / (\beta + 1) \ll 1/g_m$.

Key Design Trade-offs

Choice	Benefit	Cost
Lower $1/g_m$	Better impedance match	More power ($I_C = g_m V_T$)
Larger R_E	Less signal shunting	More voltage headroom used
Larger C_B	Better base bypassing	Larger chip area (ICs)
Smaller R_S	Less gain attenuation	May not be controllable

“The CB stage is the CE stage’s twin — same gain magnitude, same output impedance, opposite sign. Its low input impedance is a weakness at low frequencies but a superpower at high frequencies.” — Razavi